

ExamNotes AI

Importance: +P+P+P

Sub Topics

+P Topics:

- BST Self-Balancing Concepts
- Threaded Binary Trees

+P+P Topics:

- BST Operations: Insertion and Deletion
- BST Traversal Algorithms

+P+P+P Topics:

- Binary Search Tree Properties and Search Complexity
- AVL Trees and Height Balancing

Notes

Binary Search Tree (BST) Core Summary

Definition: Node-based binary tree where $\text{left.key} < \text{root.key} < \text{right.key}$ holds recursively.

Search Complexity:

Average Case: $O(\log n)$

Worst Case: $O(n)$ (when tree degrades to a linked list)

Space Complexity: $O(n)$ space required for storing nodes.

Traversal Techniques

In-Order (Left, Root, Right): Yields elements in sorted ascending order.

Pre-Order (Root, Left, Right): Used for copy operations and serialization.

Post-Order (Left, Right, Root): Used for tree deletion and bottom-up evaluation.

Level-Order (Breadth-First): Processes nodes level by level using a Queue.

Key Operations

Search: Binary search decision at each node; move left if key is smaller, right if larger.

Insert: Traverses to find an empty spot maintaining the BST invariant.

Delete: Three cases: node with no child (leaf), node with one child, and node with two children (replace with in-order predecessor or successor).

Self-Balancing Binary Search Trees

AVL Tree: Strict balance condition where balance factor = $|\text{height}(\text{left}) - \text{height}(\text{right})| \leq 1$.

Red-Black Tree: Weakly balanced tree using node colors to ensure $O(\log n)$ operations.

Revision Points

- BST Rule: Left child $<$ Parent $<$ Right child.
- In-order traversal of a BST always yields sorted data.
- Average time complexity for Search, Insert, and Delete is $O(\log n)$.
- Worst-case time complexity is $O(n)$ for skewed BSTs.

- Deletion of a node with two children requires replacing it with its In-order Successor or Predecessor.
- AVL tree height difference balance factor must be in $\{-1, 0, 1\}$.
- Height of a balanced BST with n nodes is $O(\log n)$.

Important Questions

Short Questions:

- [object Object]
- [object Object]
- [object Object]
- [object Object]

Long Questions:

- [object Object]
- [object Object]

Diagram Question:

[object Object]

Diagram

Mermaid Diagram Definition:

```
graph TD
  A[Root Node] --> B[Left Subtree]
  A --> C[Right Subtree]
  B --> D[Smaller Keys]
  C --> E[Larger Keys]
```

Charts

Worst-Case Time Complexity Operations (N units)

Balanced Search: 3
 Balanced Insert: 3
 Skewed Search: 10
 Skewed Insert: 10